

Data Analyst Capstone Project 2026

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EXECUTIVE SUMMARY



Technology Landscape

JavaScript remains the dominant language (79% adoption), while TypeScript and Rust show the strongest growth momentum. Legacy technologies like Cobol and Fortran face declining interest.

Job Satisfaction

Developers report an average satisfaction of 7.15/10, with career satisfaction and autonomy being the strongest drivers. Remote workers are significantly more satisfied than in-person counterparts.

Compensation

Median annual compensation is \$65,858, with US developers earning the highest. Remote work correlates with higher pay across most countries.

Geography & Age

Python shows strong adoption in India and the UK. Younger developers (18-34) favor TypeScript, Python, and Rust, while older developers stick with Java and C#.

AI & Learning

AI sentiment is cautiously optimistic, with younger developers more open to adoption. Coding bootcamps produce competitive compensation outcomes vs traditional education.

INTRODUCTION

Background

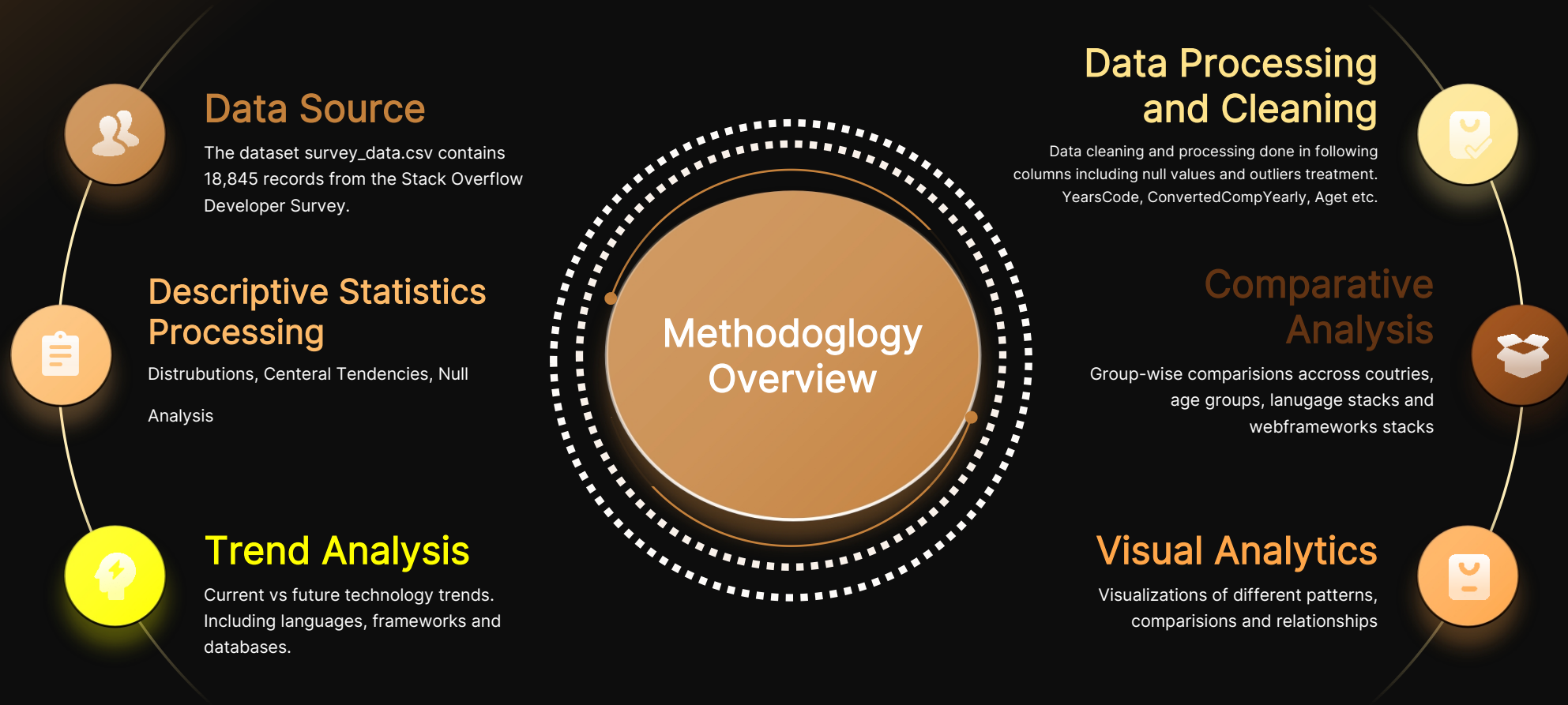
The dataset comes from the annual Stack Overflow Developer Survey, one of the largest and most comprehensive surveys of software developers globally. The survey captures information on demographics, technology usage, employment, compensation, and attitudes toward emerging trends.

Objectives

This analysis aims to:

- Understand the current technology landscape and identify emerging trends
- Analyze job satisfaction patterns and their key drivers
- Examine compensation distribution across demographic and professional segments
- Map geographic technology adoption patterns
- Identify age-related differences in technology preferences
- Explore relationships between work patterns, AI attitudes, learning pathways, and professional outcomes

METHODOLOGY

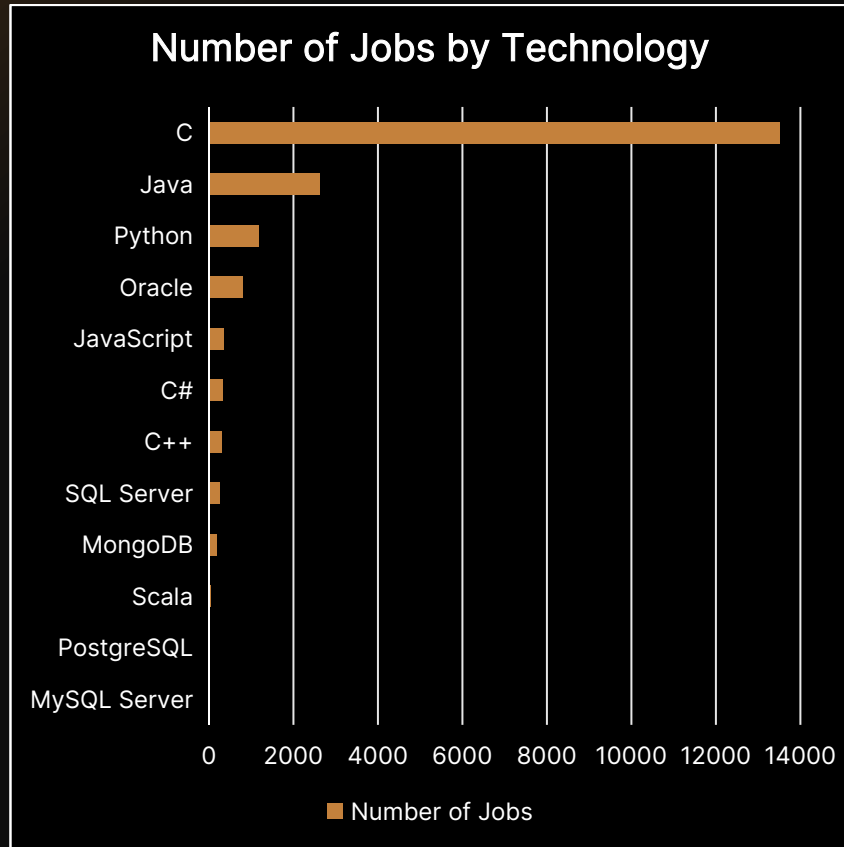


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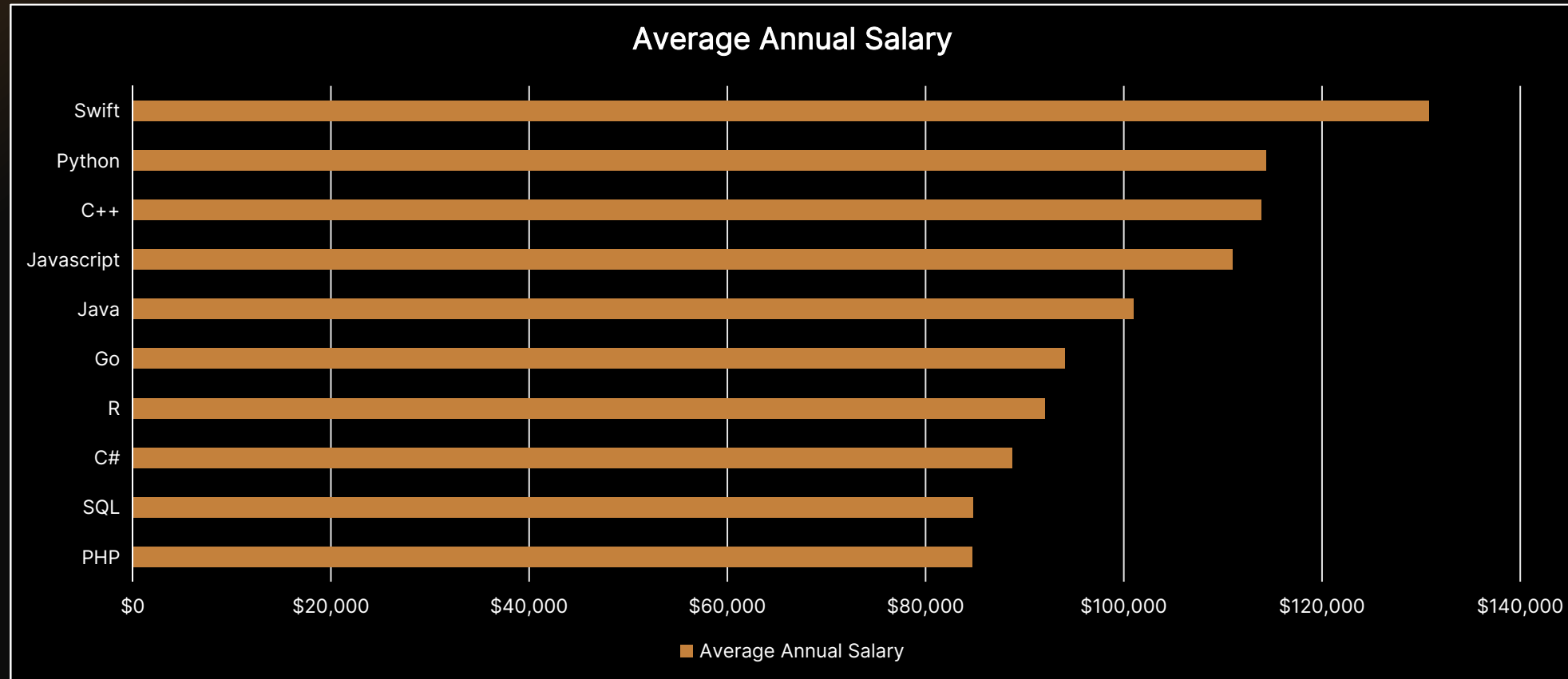
Results



JOB POSTINGS



POPULAR LANGUAGES

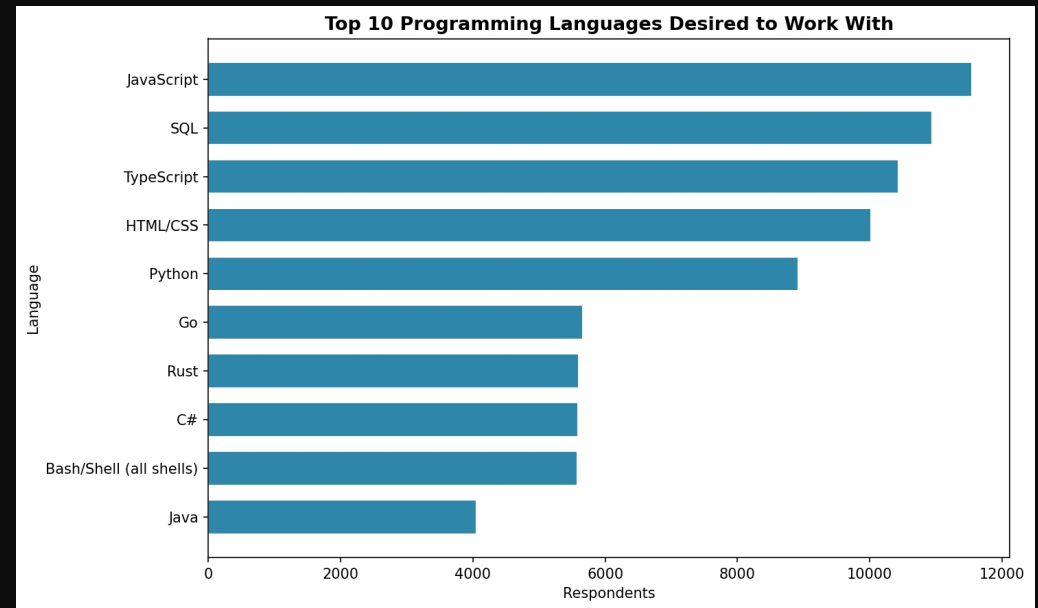
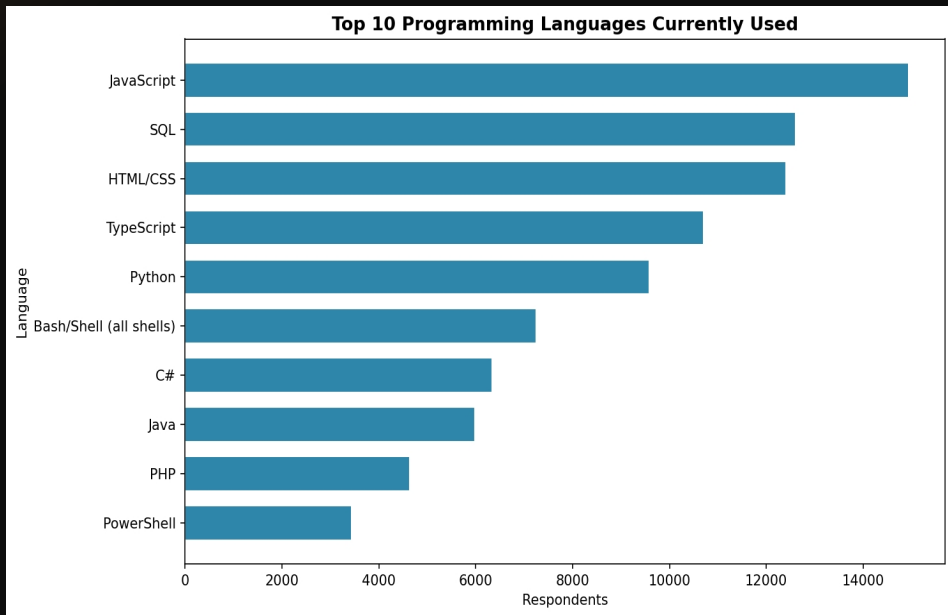


PROGRAMMING LANGUAGE TRENDS

Summarize key trends shown in the charts.

Current Year

Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Key Findings:

- TypeScript is the highest-growth language (~57% current vs. ~27% wanted), already surpassing Java/C# in current usage and topping the future "wanted" list alongside JavaScript.
- JavaScript is the universal baseline (~79% current vs. ~27% wanted), dominating current adoption as a foundational requirement but seeing a lower "want" percentage as it is now considered a standard, "solved" skill.
- Python maintains sustained mainstream demand (~51% current vs. ~24% wanted), holding a strong top-5 position today while continuing to rank second for future interest.
- Rust and Go represent the upcoming "gap" languages (<10% current vs. ~7% and ~9% wanted), currently sitting outside the top 10 for usage but breaking into the top tier for future developer desire.

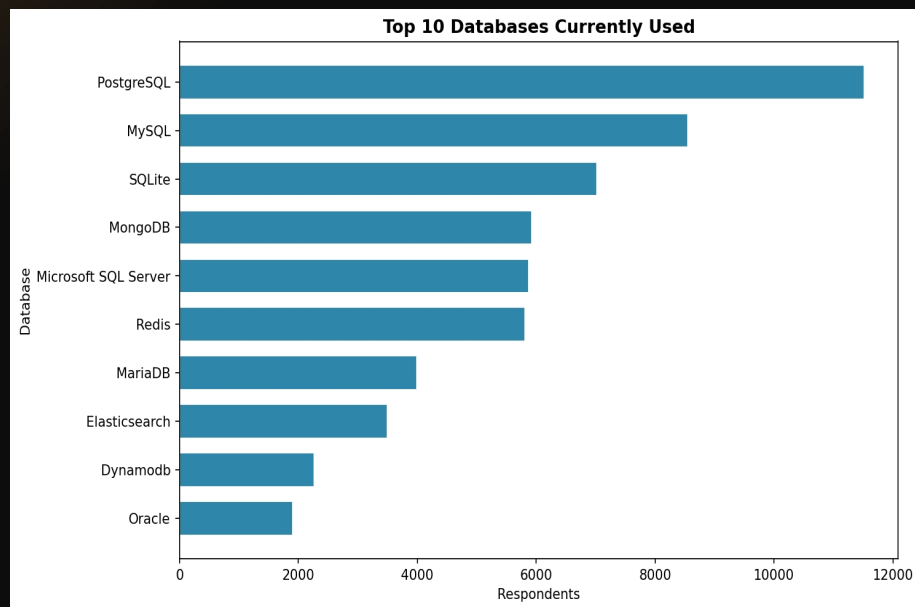
Implications:

- TypeScript offers a longer growth runway than JavaScript; while JavaScript ensures immediate, broad employability, TypeScript represents the single best language for long-term career development.
- Python's ecosystem is expanding rather than stabilizing; its current strength in data science is being actively sustained and propelled into the future by the massive growth of AI/ML.
- Rust and Go present higher niche career upside than established languages; while foundational languages like SQL and HTML/CSS are essential to know, Rust and Go offer developers the biggest competitive advantage due to low current supply and high future demand.

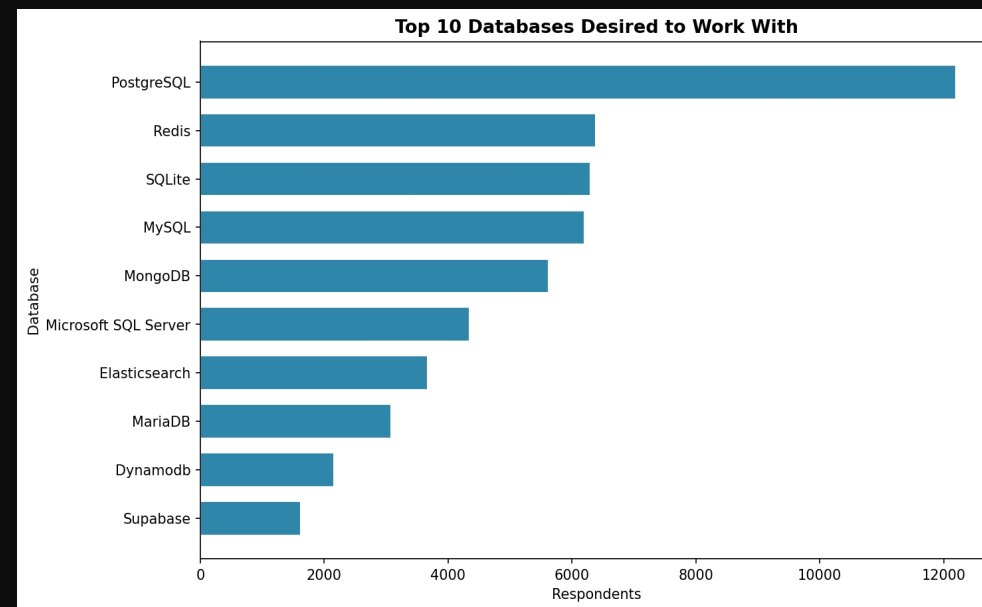
DATABASE TRENDS

Summarize key trends shown in the charts.

Current Year



Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Key Findings:

- PostgreSQL Dominance: PostgreSQL is currently the most used database due to its open-source nature, strong feature set, and enterprise adoption; it also tops the future "most wanted" list, confirming its continued growth trajectory.
- MySQL Decline: MySQL currently ranks third but has been losing ground to PostgreSQL in recent years, dropping significantly in future "want" rankings as developers actively move away.
- MongoDB & NoSQL Relevance: MongoDB leads the NoSQL category, confirming its place as the default document database, and ranks second in future desire, showing sustained interest beyond its current adoption.
- Emerging Analytics (DuckDB & ClickHouse): While Redis and Elasticsearch currently lead specialized usage in caching and search, DuckDB has entered the future top 10 despite low current usage, and ClickHouse is rising rapidly as a column-oriented database for analytics workloads.
- Cloud vs. On-Premise Enterprise: SQL Server remains relevant today in enterprise .NET environments, but future desire is shifting heavily toward cloud databases (DynamoDB, BigQuery, Firebase), reflecting ongoing cloud migration trends.
- SQLite Ubiquity: SQLite currently ranks second, driven by its ubiquity in mobile, embedded, and local development environments.

Implications:

- Core Skill Investments: PostgreSQL expertise is the most valuable database skill for broad employability today, and alongside MongoDB, it represents the safest skill investment for the next 3–5 years.
- The MySQL Pivot: MySQL knowledge is actively depreciating; existing MySQL users should prioritize learning PostgreSQL to stay aligned with market shifts.
- Analytics Engineering Opportunities: DuckDB and ClickHouse represent high-value, early-stage opportunities for specialization in analytics engineering.
- Cloud Ecosystem Integration: Cloud-native databases are growing fast, meaning modern cloud skills must complement traditional database skills to maximize career relevance.
- Baseline Developer Skills: SQLite's high rank confirms that local and embedded database skills remain a fundamental requirement for every developer, regardless of role.

02

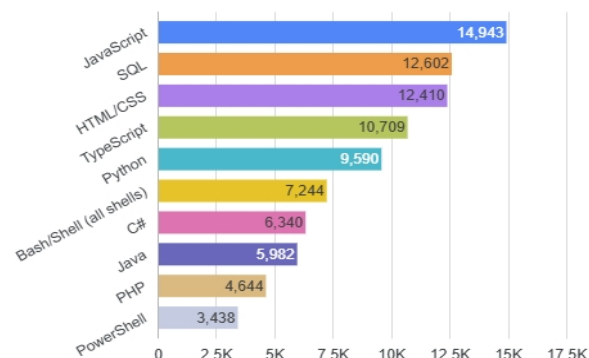
DASHBOARD



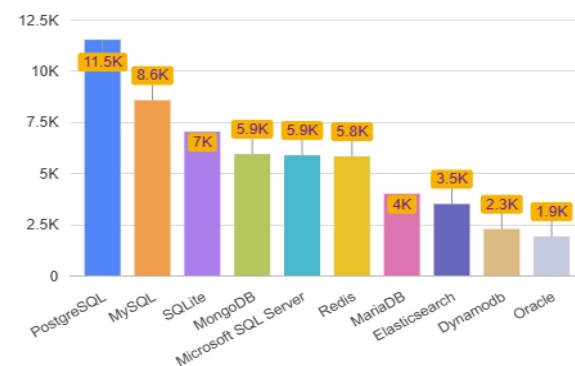
DASHBOARD TAB 1: Current Technology Usage

Current Technology Usage

Top 10 Languages Used



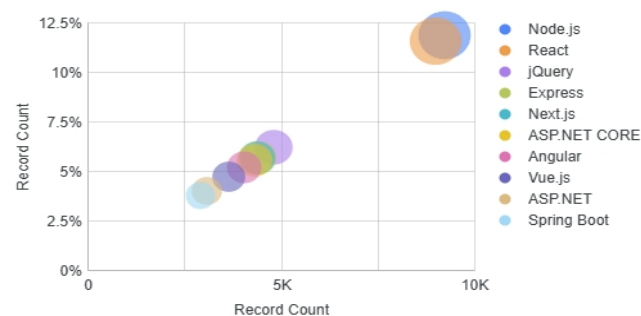
Top 10 Databases Used



Top 10 Platforms Used



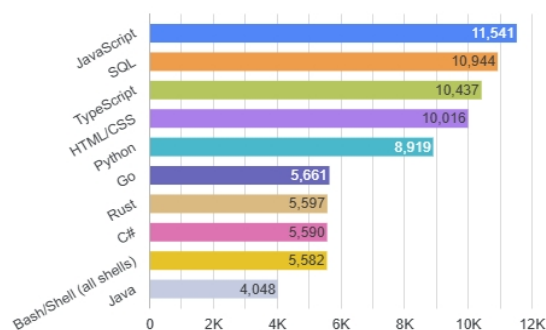
Top 10 Web Frameworks Used



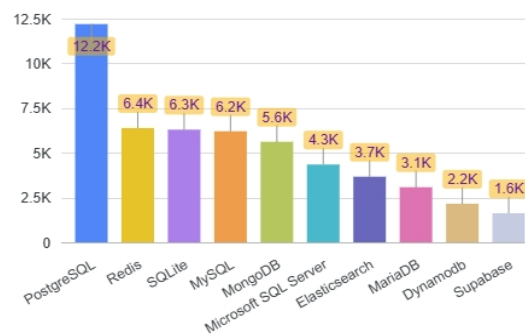
DASHBOARD TAB 2: Future Technology Trends

Future Technology Trends

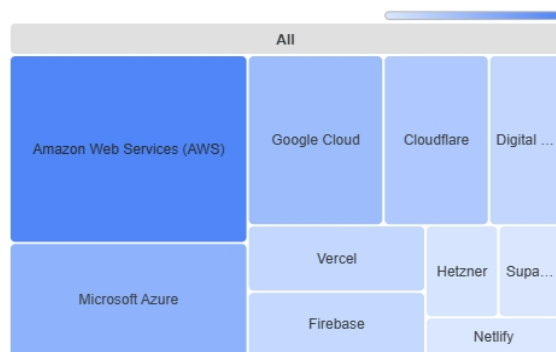
Top 10 Languages Desired Next Year



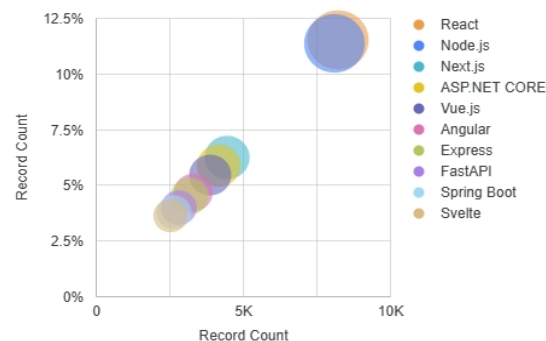
Top 10 Databases Desired Next Year



Top 10 Desired Platforms



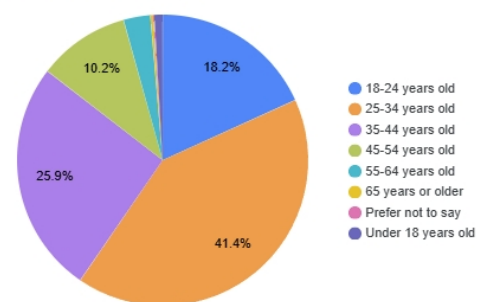
Top 10 Desired Web Frameworks



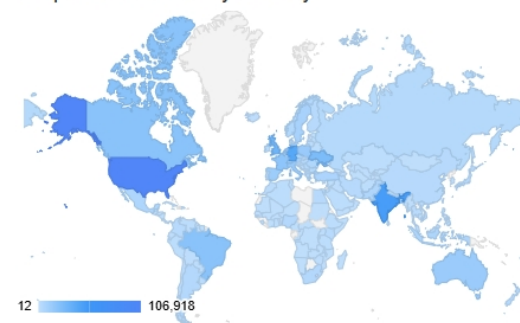
DASHBOARD TAB 3: Demographics

Future Technology Trends

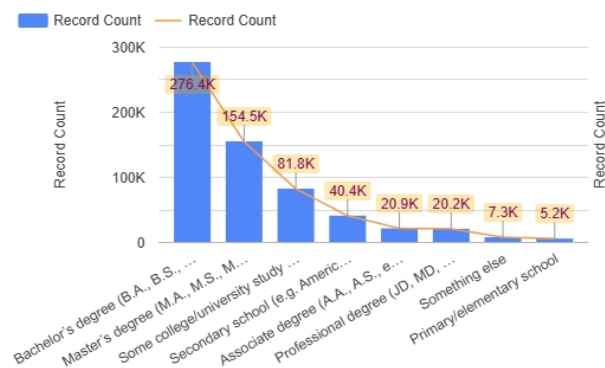
Respondents by Age



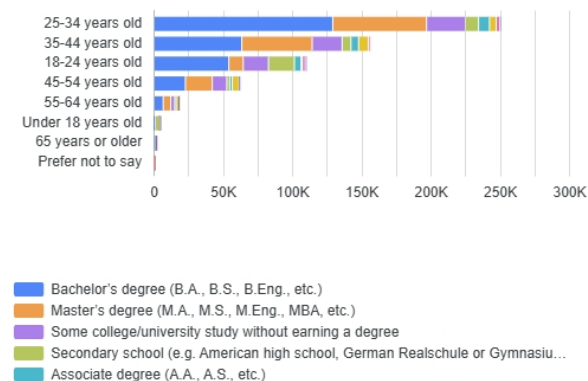
Respondents Count by Country



Respondent Distribution by Education Level



Respondent Count by Age, Classified by Education Level



DISCUSSION

Technology Trends

The technology landscape revealed by this survey confirms several well-known trends while surfacing emerging patterns. The JavaScript ecosystem continues to dominate, but the rapid rise of TypeScript — now the #1 most-wanted language — signals a qualitative shift in developer preferences toward type safety at scale. This mirrors industry trends where large codebases increasingly adopt TypeScript for maintainability.

Rust and Go represent the most significant 'adoption gap' opportunities: relatively few developers currently use them, but demand is disproportionately high. For organizations hiring, prioritizing Rust or Go skills may yield access to a smaller but highly motivated talent pool.

PostgreSQL's lead over MySQL in both current and desired usage confirms a long-anticipated tipping point. MySQL, once the default open-source relational database, is now in relative decline. DuckDB's emergence in the top 10 wanted databases despite minimal current usage is notable — it signals growing interest in embedded analytical databases, particularly among data engineers.

OVERALL FINDINGS & IMPLICATIONS

Summary of Key Findings

1. Technology Trends: JavaScript remains dominant (79% adoption), TypeScript and Rust are rising fastest (TypeScript is #1 wanted at 27%, Rust enters top 10 wanted despite not being in top 10 current). Legacy technologies (Cobol, Fortran, Perl) show declining interest. PostgreSQL has overtaken MySQL as the leading database.
2. Geography: JavaScript is ubiquitous globally (70%+ in all top-10 countries). TypeScript adoption is strongest in Western Europe. Python is particularly strong in India and the UK, driven by outsourcing and data science demand.
3. Age: Younger developers favor TypeScript, Python, and Rust. Older developers stay with C#, Java, and established ecosystems. AI tool adoption is highest in the 18-34 demographic. The experience-compensation curve shows diminishing returns after 15-20 years.

Implications & Recommendations

For Developers

- Invest in TypeScript — it offers the best risk/reward ratio for career development, with both high current demand and strong growth trajectory.
- Learn PostgreSQL if you work with databases — it is the clear market leader with sustained growth momentum.
- Consider Rust or Go if you want to differentiate yourself — demand far outstrips current supply, creating premium positioning.
- Prioritize remote-capable roles — they correlate with higher satisfaction and compensation across nearly all comparisons.

For Employers

- Support TypeScript adoption in your tech stack — it improves developer productivity and makes your company more attractive to top talent.
- Offer remote and hybrid options — the satisfaction and retention benefits are clear across all compensation levels.
- Invest in mid-career retention — the 35-44 age bracket is the satisfaction peak and represents your most productive engineers.
- Build AI-assisted development workflows — the next generation of developers expects AI tooling as part of their standard toolkit.

CONCLUSION

The 2024 Stack Overflow Developer Survey reveals a developer ecosystem in transition. The technology landscape is being reshaped by the TypeScript revolution, the PostgreSQL ascendancy, and the early but accelerating impact of AI tools. Job satisfaction remains moderate, driven primarily by autonomy and career growth rather than compensation alone. Remote work has cemented its place as a structural feature of the industry, correlating positively with both happiness and earnings.

For developers, the message is clear: invest in TypeScript, PostgreSQL, and cloud-native skills; prioritize remote-capable roles; and prepare for AI-assisted development as the new normal. For employers, the data supports investing in developer experience, offering flexible work arrangements, and modernizing technology stacks to attract and retain top talent.

The common thread across all analyses is that the industry is becoming more specialized, more distributed, and more tool-augmented. Developers who embrace these trends — by learning modern languages, working remotely, and leveraging AI tools — position themselves for the strongest career outcomes.

03

APPENDIX



Extra Charts

